CAPSTONE PROJECT REPORT

Project Title	Enterprise Banking Risk Management Analytics Solution
Track	Data Analytics
Job Profile	Professional Data Analysis
Program Name	Nano Degree
Team Name	Risk Wise
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Declaration and Certificate of Originality

We, the undersigned team members, declare that this capstone project report is our original work and has been prepared for academic evaluation purposes. Any external datasets, tools, libraries, frameworks, or resources used in the project are acknowledged within this final project documentation according to the committee requirements.

We confirm that the proposed project idea, implementation plan, and expected deliverables are developed by the team and that the project will follow ethical, legal, and professional standards related to data, security, software development, and technology usage.

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Dedication

We dedicate this project to our families, instructors, mentors, and colleagues whose encouragement and support enabled us to complete this capstone. We also dedicate it to aspiring data professionals who seek to apply analytics responsibly to real business challenges.

Acknowledgement

The team would like to acknowledge the support, guidance, and feedback provided by the program management, supervision committee, mentors, academic team members, and any external contributors who supported the project development process.

We extend our sincere appreciation to Prof. Dr. Rania El Gohary, Dr. Ahmed Zakaria, Dr. Yasser Ali, and Dr. Wael Mahrous of NXT for their academic guidance, technical feedback, and continued support throughout the project.

Abstract

Banking risk analysis requires consistent integration of loan, customer, payment, collection, collateral, default, recovery, branch, and loan-officer data. This capstone develops an Enterprise Banking Risk Management Analytics Solution that converts fragmented records into a governed decision-support model. SQL Server organizes the data through Bronze, Silver, and Gold layers. Six Gold views provide three shared dimensions and three fact-oriented analytical structures. Power BI consumes these views through a dimensional semantic model supported by a DAX Calendar and a centralized _Measures table. The current validated Gold outputs contain 30,000 loan records, 97,642 cashflow events, and 92,932 risk events, with zero reported amount differences and zero missing customer, product, or organization keys. The report follows four distinct management pages: Loan Portfolio Overview, Credit Risk and Default Analysis, Collections and Recovery, and Branch and Loan Officer Performance. Predictive default modeling, Probability of Default, Explainable AI, early warning, and Expected Credit Loss remain governed future extensions rather than operational outputs in the current implementation.

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List of Abbreviations

Abbreviation	Meaning
BI	Business Intelligence
KPI	Key Performance Indicator
SQL	Structured Query Language
DAX	Data Analysis Expressions
EDA	Exploratory Data Analysis
ML	Machine Learning
DPD	Days Past Due
NPL	Non-Performing Loan
PD	Probability of Default
LGD	Loss Given Default
EAD	Exposure at Default
ECL	Expected Credit Loss
IFRS 9	International Financial Reporting Standard 9
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
SHAP	SHapley Additive exPlanations

Abbreviations are limited to terms used in the current implementation or its explicitly defined predictive roadmap.

List of Symbols

Symbol	Meaning / Description
N_L	Total number of loans in the selected filter context
N_D	Number of defaulted loans in the selected filter context
A_due	Total amount contractually due
A_paid	Total amount actually paid
DPD	Days Past Due for a payment or risk event

The symbols support the documented ratios: Default Rate = N_D / N_L and Collection Rate = A_paid / A_due. Future ECL analysis may use ECL = PD x LGD x EAD after governed inputs are available.

1. Executive Summary

1.1 Project Overview

The Enterprise Banking Risk Management Analytics Solution is an end-to-end analytical project designed to convert fragmented lending and risk-related data into a consistent management view of portfolio performance. The solution focuses on loan portfolio composition, credit risk, default behavior, payment and collection performance, recovery activity, collateral support, and organizational performance across branches and loan officers.

The project was structured around a controlled data pipeline rather than direct reporting from raw source files. Source data was first ingested into a Bronze layer, then cleaned and standardized in a Silver layer, and finally transformed into a Gold analytical layer designed specifically for business intelligence. This separation improves traceability, data quality, reuse, and consistency of calculations across the reporting solution.

1.2 Current Implementation Status

The Bronze, Silver, and Gold SQL stages are complete and validated. The final Gold layer contains six views: gold dim_customer, gold dim_product, gold dim_organization, gold fact_loans, gold fact_cashflow, and gold fact_risk_events. These views have been imported into Power BI. The Calendar table, centralized _Measures table, and one-to-many dimension-to-fact relationships are established. Dashboard 1 and Dashboard 2 are in visual refinement, while the Collections and Branch/Officer pages follow the same approved KPI-to-chart design. Machine-learning and ECL outputs are documented as future extensions until they are fully governed, validated, and integrated.

Gold View	Analytical Role
gold dim_customer	Customer attributes and credit-related profile information
gold dim_product	Loan product classification and product-level analysis
gold dim_organization	Branch and loan officer organizational analysis
gold fact_loans	Loan portfolio and approved exposure analysis
gold fact_cashflow	Due, paid, late-payment, collection and cashflow analysis
gold fact_risk_events	Defaults, recovery and other risk-event analysis

Design Principle

Power BI consumes the Gold analytical views rather than querying raw or partially cleaned data. This keeps business reporting separated from data-cleaning logic and supports reproducible KPI results.

1.3 Analytical Storyline

The reporting layer follows one non-repetitive management storyline. Dashboard 1 explains portfolio size, approved exposure, product mix, lending trend, and collateral composition. Dashboard 2 evaluates realized default risk through Default Rate, Defaulted Loans, Default Amount, Average DPD, and risk-event behavior. Dashboard 3 measures Due Amount, Paid Amount, Collection Rate, late-payment behavior, and Recovery Amount. Dashboard 4 compares workload, approved exposure, and collection activity across branches and loan officers. Each KPI is defined once and each supporting chart explains its product, customer, organization, or time driver.

Dashboard	Primary Management Question
Loan Portfolio Overview	How large is the portfolio and how is lending distributed?
Credit Risk & Default Analysis	Where did credit risk and default occur, and what factors explain it?
Collections & Recovery	How much was due, paid and recovered, and where are collection gaps?
Branch & Loan Officer Performance	Where is activity concentrated and who manages the largest workload?

1.4 Key Business Value

- A single analytical view of lending, risk, collections and organizational performance.
- Consistent KPI definitions supported by SQL-to-Power BI reconciliation.
- Clear separation between descriptive portfolio metrics and diagnostic risk drivers.
- Improved visibility into default concentration, delinquency, late payments and recovery.
- A reusable dimensional model that supports filtering by customer, product, organization and time.
- A scalable foundation for future machine-learning outputs such as Probability of Default and customer risk segmentation.

1.5 Executive Outcome

The project therefore moves the analysis from disconnected records toward a structured risk decision-support environment. The value is not limited to dashboard visualization: the underlying Bronze, Silver and Gold architecture provides a repeatable analytical foundation, while the Power BI model translates that foundation into business questions, measurable KPIs, explanatory visuals and management insights.

2. Problem Statement

2.1 Business Context

Banking risk management depends on the ability to connect lending activity with customer characteristics, payment behavior, collateral, default events, recovery activity and organizational ownership. In practice, these elements are commonly stored in different tables or files, with different levels of granularity and different update patterns. A loan can be linked to a customer, product, branch and officer, while its payment and risk history may generate many related events over time.

When these datasets are reviewed separately, management can obtain different answers to the same business question depending on the source used. A portfolio total calculated from a loan table may not align with a collection table if joins duplicate records. Default counts can also be distorted when event-level tables are joined directly to loan-level data without controlling the analytical grain.

2.2 Core Business Problem

The core problem addressed by the project is the absence of one governed analytical structure that brings the available banking datasets together without losing their original business grain. The project must preserve loan-level exposure, transaction-level cashflow activity and event-level risk behavior while still allowing the user to analyze all three from a shared set of customer, product, organization and date dimensions.

2.3 Management Consequences

- Difficulty obtaining a consistent view of total portfolio size and approved exposure.
- Limited ability to isolate products, branches or customer profiles associated with higher default risk.
- Potential duplication of balances when one-to-many tables are joined without a dimensional model.
- Poor visibility into the gap between amounts due and amounts actually paid.
- Limited ability to distinguish historical default analysis from collection and recovery performance.
- Reporting effort spent reconciling numbers instead of interpreting business drivers.

Analytical Requirement

The solution must keep different business grains separate, connect them through reusable dimensions, and ensure that KPIs can be validated back to SQL before they are used for management reporting.

2.4 Portfolio and Risk Questions

- What is the total number of loans and the total approved lending amount?
- How is the portfolio distributed by loan type and product?
- What percentage of loans are supported by collateral?
- What proportion of the portfolio has defaulted, and what is the financial value of those defaults?
- Which customer, product and organizational characteristics are associated with the highest risk?
- How severe is delinquency as measured through Days Past Due and recorded risk events?

2.5 Collection and Recovery Questions

- How much was contractually due compared with the amount actually paid?
- What is the portfolio Collection Rate and how does it vary by method, product, branch or period?
- Which products contain the highest concentration of late payments?
- How much has been recovered from risky or defaulted exposures?
- Which risk categories contribute most to recovery activity?

2.6 Organizational Questions

- How is loan volume distributed across branches?
- Which loan officers manage the largest number or value of loans?
- How does collection activity differ across branches?

- Where should management focus further investigation based on risk, workload or collection results?

2.7 Problem Definition Summary

The project is therefore not simply a visualization exercise. It is a data integration and analytical modeling problem in which reliable reporting depends on controlling data quality, defining grain, preventing duplicated aggregation, and using consistent business calculations. The reporting solution must answer management questions at both summary and diagnostic levels while maintaining traceability to the underlying data.

3. Project Objectives

3.1 Primary Objective

The primary objective is to build a reliable banking risk analytics solution that integrates the available lending, payment and risk datasets into a governed analytical model and translates them into decision-oriented Power BI dashboards.

Objective Area	Specific Objectives
Data	Profile source data; identify quality issues; standardize types, categories and keys; preserve traceability across layers.
SQL Architecture	Implement Bronze, Silver and Gold layers and create reusable business-ready Gold views.
Modeling	Create a dimensional Power BI model using shared dimensions and separate fact grains.
KPIs	Define measurable portfolio, default, delinquency, collection, recovery and organizational indicators.
Analysis	Move from descriptive results to diagnostic analysis of products, customers, branches, officers and time trends.
Visualization	Build interactive dashboards where each chart explains or decomposes a relevant KPI.
Validation	Reconcile counts and amounts between SQL and Power BI and test relationship/filter behavior.
Decision Support	Highlight risk concentration, collection gaps and operational areas requiring management attention.

3.2 Success Criteria

- Source-to-Gold transformations are traceable and repeatable.
- Gold views contain no missing dimension keys after validation.
- Loan, cashflow and risk-event totals reconcile to the validated SQL outputs.
- Power BI calculations respond correctly to business filters and date context.
- Each dashboard answers a distinct set of analytical questions without unnecessary KPI duplication.
- The final report documents business definitions, technical design, validation and future extensions.

4. Project Scope

4.1 In Scope

Area	Included Activities
Data Understanding	Source identification, profiling, grain review, key-field review and business-field interpretation.
Data Quality	Null, duplicate, type, range, category, key and referential-integrity checks.
Data Preparation	Cleaning, standardization, transformation, integration and quality validation in SQL.
Data Architecture	Bronze raw layer, Silver trusted layer and Gold business-ready analytical layer.
Gold Model	Three dimensions and three fact-oriented analytical views.
Power BI	Calendar table, relationships, measures, KPIs, slicers and analytical dashboard pages.
Analysis	Portfolio, default, DPD, risk-event, collection, recovery, collateral, branch and officer analysis.
Testing	Row-count validation, amount reconciliation, dimension-key checks, KPI testing and user-facing validation.

4.2 Out of Scope

- Replacement of the bank core banking or transaction-processing platform.
- Automatic approval or rejection of loan applications.
- Execution of live banking transactions or modification of customer balances.
- Full regulatory IFRS 9 Expected Credit Loss calculations without all required regulatory inputs, assumptions and governance.
- Production-grade deployment of machine-learning models in the current implementation.
- Automated regulatory filing or replacement of formal risk/compliance systems.

4.3 Scope Control Principle

The report distinguishes between functionality already implemented or designed in the current analytics solution and future enhancements. Predictive default modeling, Probability of Default, segmentation, Explainable AI and ECL extensions are treated as future work unless and until they are fully developed, validated and integrated into the reporting model.

5. Target Users and Stakeholders

5.1 Stakeholder Matrix

Stakeholder	Primary Need	Typical Decision
Senior Management	Portfolio health, exposure, risk concentration and strategic trends	Where should management focus attention or adjust portfolio strategy?
Risk Management	Defaults, delinquency, exposure, risk events and high-risk segments	Which products, customers or branches require deeper risk review?
Credit Department	Loan quality, customer profile and product performance	How is credit quality changing across lending segments?
Collections Team	Due amounts, payments, collection rate, late payments and recovery	Which cases or segments should be prioritized for collection activity?
Branch Management	Branch lending, default and collection performance	How does the branch compare with peers and where are operational gaps?
Loan Officers	Assigned loan volume and exposure	How is workload and managed lending distributed?

5.2 Management User Perspective

The dashboard is designed to allow a senior user to begin with a high-level KPI and then investigate its drivers. For example, a Default Rate card provides the overall result, while supporting charts show how that rate differs by loan type, credit score, branch or risk status. This approach reduces the need to search across disconnected reports and makes the analytical path more intuitive.

User-Centered Design

The solution is structured around management questions rather than around source-table names. Users see portfolio, risk, collection and organizational views, while the technical complexity remains inside the data model.

5.3 Information Requirements

User Group	Required Indicators / Views
Senior Management	Total Loans, Approved Amount, Default Rate, Default Amount, Collection Rate, Recovery Amount, branch comparisons and major trends.
Risk Management	Defaulted Loans, Average DPD, Total Risk Events, risk status, credit score bands, customer risk attributes and exposure concentration.
Collections	Due Amount, Paid Amount, Collection Rate, Late Payment Rate, payment method and recovery by risk status.
Branch Management	Loan volume, approved amount, default exposure, collection amount, branch ranking and officer workload.
Data / BI Team	Row counts, keys, relationships, DAX logic, SQL reconciliation, refresh behavior and model performance.
Compliance / Risk Control	Defined KPI logic, exception visibility, traceability, access controls and documentation of assumptions.

5.4 Stakeholder Engagement

Stakeholder engagement is important during KPI definition, dashboard design and validation. Business users determine whether a metric is meaningful, while the technical team verifies that its calculation is supported by the available data and the defined grain. This separation prevents a visually attractive report from presenting a technically incorrect or business-ambiguous number.

5.5 Stakeholder Decision Flow

1. Observe the overall KPI or trend.
2. Filter by date, product, branch, customer characteristics or risk status.
3. Use supporting charts to identify concentration or variance.
4. Drill into the business segment that explains the KPI movement.
5. Use the insight to prioritize monitoring, collection follow-up or management review.

6. Proposed Solution

6.1 Solution Overview

The proposed solution uses a layered SQL architecture followed by a Power BI semantic model. The objective is to ensure that raw source complexity is resolved before the data reaches the dashboard. Each layer has a defined responsibility: Bronze preserves source data, Silver creates trusted standardized data, Gold presents business-ready analytical structures, and Power BI applies semantic relationships, DAX measures and interactive visualizations.

6.2 End-to-End Flow

Step	Layer / Component	Purpose
1	Source Data	Banking files/tables containing customer, loan, product, organization, payment, default and related information.
2	Bronze	Raw ingestion and source reconciliation; minimal transformation beyond technical loading.
3	Silver	Cleaning, standardization, type conversion, duplicate handling, domain validation and key checks.
4	Gold	Six analytical views organized around dimensions and fact grains.
5	Power BI Model	Calendar table, relationships and business semantic structure.
6	Measures & KPIs	Portfolio, default, DPD, collection, recovery and organizational calculations.
7	Dashboards	Interactive portfolio, risk, collection and branch/officer analytical pages.
8	Business Insight	Interpretation of concentration, trends, drivers and areas requiring management attention.

6.3 Design Philosophy

The solution follows a separation-of-concerns principle. Data quality rules are implemented as close as possible to the SQL preparation layer, while business measures are defined centrally in Power BI. This avoids repeating cleaning logic inside visuals and makes the final dashboard easier to maintain and explain.

6.4 Dimensional Analytical Solution

The Gold layer and Power BI model separate descriptive dimensions from measurable business events. The shared tables gold dim_customer, gold dim_product, and gold dim_organization filter gold fact_loans, gold fact_cashflow, and gold fact_risk_events. Calendar provides date context, while _Measures stores reusable DAX measures and has no relationship of its own. This Galaxy-style design avoids direct fact-to-fact relationships and prevents duplicated aggregation across different business grains.

6.5 KPI-to-Chart Design

A central principle of the visualization design is that each major chart must support a defined analytical question. KPI cards answer what happened at a summary level. Supporting charts explain where, when or why the result occurred. This is especially important for credit-risk reporting because a headline rate alone does not identify the concentration or drivers behind the risk.

KPI	Supporting Diagnostic View	Management Interpretation
Default Rate %	Default Rate by Loan Type / Credit Score / Branch	Which segments are driving historical default?
Collection Rate %	Collection Rate by Method / Product / Period	Where is the collection gap strongest?
Average DPD	Average DPD by Risk Status	How severe is delinquency across risk categories?
Total Approved Amount	Approved Amount Trend and Product Mix	How is lending volume changing and where is exposure concentrated?
Collections per Branch	Collection Amount by Branch	Which branches manage the largest collection activity?

6.6 Expected Management Experience

The final user experience is intended to be progressive: start with a portfolio summary, move to historical risk, examine collections and recovery, then compare organizational performance. Filters maintain context across visuals so that a manager can move from a bank-level view to a branch, product or customer segment without recalculating data manually.

7. System Features

7.1 Loan Portfolio Analysis

- Total number of loans in the portfolio.
- Total approved lending amount.
- Average loan amount.
- Loan distribution by product or loan type.
- Approved amount trend over time.
- Collateralized versus non-collateralized portfolio composition.

7.2 Credit Risk and Default Analysis

- Default Rate as the share of loans that defaulted.
- Number of defaulted loans.
- Total financial amount associated with defaulted loans.
- Average Days Past Due as a delinquency severity measure.
- Total risk-event volume and trend over time.
- Default concentration by product, credit score and branch.
- Risk-status comparison for delinquency and event behavior.

7.3 Customer Risk Driver Analysis

Customer attributes are used as explanatory dimensions rather than as a separate dashboard that duplicates risk information. Relevant attributes include credit score, annual income, employment status, credit history and debt-to-income information where available. These characteristics allow the dashboard to explain why the same portfolio-level risk rate may differ substantially across customer groups.

Feature Design Rule

Historical default and risk measures are kept on the Credit Risk page rather than repeated on the Portfolio page. This preserves a clean analytical story and prevents users from seeing slightly different versions of the same KPI in multiple places.

7.4 Collections and Payment Analysis

- Total amount due for payment.
- Total amount paid.
- Collection Rate as Paid Amount divided by Due Amount.
- Late Payment Rate.
- Due versus Paid comparison by month or selected period.
- Collection performance by payment or event method.
- Late-payment concentration by product.

7.5 Recovery Analysis

- Total Recovery Amount from recorded risk and recovery events.
- Recovery distribution by risk status.
- Comparison of recovery behavior across portfolio segments.
- Ability to connect recovery results to branch, product and customer filters.

7.6 Branch and Loan Officer Analysis

- Total number of branches represented in the model.
- Total number of loan officers.
- Loan volume by branch.
- Loans managed by loan officer.
- Average loans per officer.
- Approved lending amount by officer.
- Collection activity by branch.

7.7 Interactive Features

The Power BI report uses slicers and cross-filtering to support investigation by date, product, organization and customer-related characteristics. The objective is to maintain analytical context while the user moves between a high-level KPI and the segment that explains the result.

KPI	Definition / Business Meaning	Primary Page
Total Loans	Count of loans represented in the loan fact.	Portfolio
Total Approved Amount	Total value of approved lending.	Portfolio
Average Loan Amount	Average approved amount per loan.	Portfolio
Collateralized Loan %	Share of loans supported by collateral.	Portfolio
Default Rate %	Defaulted Loans divided by Total Loans.	Credit Risk
Defaulted Loans	Count of loans identified as defaulted.	Credit Risk
Total Default Amount	Financial value associated with defaulted loans.	Credit Risk
Average DPD	Average Days Past Due for delinquent/risk activity.	Credit Risk
Total Risk Events	Count of recorded risk events.	Credit Risk
Total Due Amount	Total amount contractually due.	Collections
Total Paid Amount	Total amount actually paid.	Collections
Collection Rate %	Total Paid Amount divided by Total Due Amount.	Collections
Late Payment Rate %	Share of payment obligations classified as late.	Collections
Total Recovery Amount	Total amount recovered from recorded recovery activity.	Collections
Total Branches	Distinct branch count.	Organization
Total Loan Officers	Distinct officer count.	Organization
Loans per Officer	Average or ratio of loan volume to officer count.	Organization
Approved Amount per Officer	Approved exposure managed by officer.	Organization
Collections per Branch	Collection activity by branch.	Organization

KPI Governance

Each KPI should be calculated once, documented, validated against SQL and reused across visuals. A chart may decompose a KPI by a dimension, but it should not redefine the KPI logic.

8. Technical Approach

8.3.1 Gold Layer Purpose

The Gold layer is the business-ready analytical layer. Instead of exposing every cleaned source table to Power BI, the project consolidates the required analytical information into six views. This keeps the Power BI model understandable, reduces unnecessary columns, and creates a stable interface between SQL preparation and business reporting.

View	Type	Primary Purpose
gold dim_customer	Dimension	Reusable customer and credit-profile attributes.
gold dim_product	Dimension	Reusable product and loan-type attributes.
gold dim_organization	Dimension	Branch and loan officer organizational attributes.
gold fact_loans	Fact	Loan portfolio, approved exposure and loan-level analysis.
gold fact_cashflow	Fact	Due, paid, late-payment and collection-event analysis.
gold fact_risk_events	Fact	Default, delinquency, recovery and risk-event analysis.

8.3.2 Gold Modeling Decisions

- Use a compact Star/Galaxy-style analytical structure.
- Keep different fact grains separate rather than building one very wide table.
- Expose only columns needed for business analysis and relationship keys.
- Use shared customer, product and organization dimensions for filtering.
- Create the Calendar dimension in Power BI instead of adding a seventh SQL Gold date view.
- Keep SQL logic simple enough to be maintainable and explainable while preserving correctness.

8.3.3 Gold Outcome

The six views become the only SQL objects required by the Power BI report. This provides a controlled hand-off between data engineering and business intelligence and makes validation significantly easier because each analytical subject has a clear source.

8.3.4 Semantic Model Design

The six Gold views are imported into Power BI and supplemented by Calendar and _Measures. Relationships use the unique dimension keys customer_id, product_id, and loan_officer_id on the one side and the matching foreign keys in the three facts on the many side. Branch attributes remain inside gold dim_organization and filter the facts through loan_officer_id. Calendar connects to the relevant fact date for each analytical page. Filtering is single-direction from dimensions to facts, and there are no direct fact-to-fact relationships.

Model Component	Role
Dim Customer	Filters loan, cashflow and risk analysis by customer characteristics.
Dim Product	Filters portfolio and risk results by product or loan type.
Dim Organization	Filters results by branch and loan officer attributes.
Calendar	Provides reporting dates, month/year fields and time-intelligence context.
Fact Loans	Loan-level portfolio and exposure measures.
Fact Cashflow	Transaction/event-level due, paid and collection measures.
Fact Risk Events	Event-level default, DPD, recovery and risk measures.

8.3.5 Relationship Principles

- gold dim_customer[customer_id] filters the three fact tables through one-to-many relationships.
- gold dim_product[product_id] filters the three fact tables through one-to-many relationships.

- gold dim_organization[loan_officer_id] filters the three facts; branch attributes are analyzed through the same dimension.
- Calendar[Date] filters the relevant application, event, or transaction date; only the required date relationship is active for each analysis.
- _Measures is a disconnected organizational table for DAX measures, and the fact tables are never linked directly to one another.

Modeling Benefit

The semantic model lets one customer, product or branch selection filter multiple business processes without combining those processes into a single duplicated fact table.

8.3.6 Measure Design

Business metrics are implemented as reusable Power BI measures so that the same definition is applied consistently across cards, charts and filter contexts. Base measures are created first, followed by ratio measures. This makes calculations easier to test and explain.

Measure Group	Examples	Design Logic
Portfolio	Total Loans, Total Approved Amount, Average Loan Amount	Calculated primarily from Fact Loans at loan grain.
Collateral	Collateralized Loan %	Ratio of loans meeting the defined collateral condition to Total Loans.
Default	Defaulted Loans, Default Rate %, Total Default Amount	Historical risk measures tied to loan/default event logic.
Delinquency	Average DPD, Total Risk Events	Calculated from risk-event fields such as event days and risk status.
Collections	Total Due Amount, Total Paid Amount, Collection Rate %	Cashflow measures aggregated at payment/event grain.
Payment Quality	Late Payment Rate %	Share of records or obligations meeting the late-payment definition.
Recovery	Total Recovery Amount	Recovery event amounts from the risk-event structure.
Organization	Total Branches, Total Loan Officers, Loans per Officer	Dimension counts and workload measures.

8.3.7 Ratio Logic

Ratios are calculated from validated base measures rather than from visual-level arithmetic. For example, Collection Rate is defined as Total Paid Amount divided by Total Due Amount. Default Rate is defined as Defaulted Loans divided by Total Loans. The denominator is protected against zero or invalid context so that the report does not display misleading infinite or error values.

8.3.8 Descriptive Analysis

Exploratory analysis is performed after the semantic model and core measures are stable. The purpose is to understand distributions, trends and segment differences before finalizing dashboard visuals. The analysis does not begin with chart selection; it begins with a management question and then identifies the KPI and dimension required to answer it.

Analytical Area	Questions Explored
Portfolio	How large is the portfolio? Which products make up the exposure? How is approved lending changing over time?
Collateral	What proportion of loans is collateralized and how is secured lending distributed?
Default	What is the default rate and which products, credit-score groups or branches have higher default concentration?
Delinquency	How severe is lateness and how does Average DPD differ by risk status?
Cashflow	How do Due Amount and Paid Amount compare over time?
Collections	Which methods, products or periods show stronger or weaker Collection Rate?
Recovery	Where are recoveries concentrated and how do they relate to risk status?
Organization	How are loan volume, approved exposure and collection activity distributed across branches and officers?

8.3.9 Analytical Sequencing

The project separates descriptive analysis from diagnostic analysis. Descriptive metrics establish what happened. Diagnostic comparisons then identify which segment, branch, product, customer characteristic or time period explains the result. This sequencing helps prevent dashboards from becoming collections of unrelated charts.

8.3.10 Diagnostic Analysis Framework

Root cause analysis in this project means investigating the drivers behind material KPI values or changes. It does not claim causal proof in a statistical sense; rather, it decomposes observed outcomes across meaningful business dimensions so that management can identify concentrations that deserve further investigation.

Headline KPI	Potential Drivers	Example Diagnostic Question
Default Rate %	Product, Credit Score, Customer Profile, Branch	Which segment contributes most to the observed default rate?
Total Default Amount	Branch, Product, Exposure Size	Is the highest risk driven by many small defaults or a smaller number of large exposures?
Average DPD	Risk Status, Product, Customer Profile	Which groups exhibit the most severe delinquency?
Collection Rate %	Time, Method, Product, Branch	Where does the gap between due and paid amounts originate?
Late Payment Rate %	Product, Customer Segment, Branch	Which portfolio segment contains repeated payment delay?
Recovery Amount	Risk Status, Collateral, Product	Which risky exposures contribute most to recoveries?

8.3.11 Risk Driver Interpretation

Customer fields such as credit score, annual income, employment status, credit history and debt-to-income information can be used as explanatory slices. Organizational fields such as branch, city, officer, designation and experience can explain where activity or risk is concentrated. The purpose is to connect a management outcome to an interpretable business characteristic rather than simply display the same total in multiple formats.

8.3.12 Dashboard Objective

The Loan Portfolio Overview page answers one central question: how large is the lending portfolio and how is it composed? Default and collection metrics are intentionally excluded from this page to keep the analysis focused on portfolio structure rather than portfolio quality.

Analytical Question	KPI	Supporting Visual
What is the size of the loan portfolio?	Total Loans	Loans by Loan Type - Column Chart
What is the total approved lending amount?	Total Approved Amount	Approved Amount Trend by Month - Line Chart
What is the average loan amount?	Average Loan Amount	Average Loan Amount by Product - Bar Chart
What proportion of loans is collateralized?	Collateralized Loan %	Collateral Status / Type - Donut or 100% Stacked Bar

8.3.13 Recommended Slicers

- Date / Reporting Period
- Loan Product / Loan Type
- Branch
- Customer-related segment where relevant

8.3.14 Interpretation

The supporting visuals explain the composition behind each KPI. For example, Total Loans becomes more meaningful when decomposed by loan type, while Total Approved Amount becomes more useful when

viewed as a trend. The collateral view provides context on the structural support behind the portfolio without mixing it with historical default outcomes.

Page Boundary

Default Rate, Defaulted Loans, Default Amount, Collection Rate, Due Amount, Paid Amount and Recovery are deliberately kept off the Portfolio page to avoid duplication across the report.

8.3.15 Dashboard Objective

The Credit Risk & Default Analysis page is the main historical risk page. It answers who defaulted, where the default is concentrated, how large the risky exposure is, and what characteristics are associated with delinquency or risk events.

Analytical Question	KPI	Supporting Visual
What percentage of loans defaulted?	Default Rate %	Default Rate by Loan Type - Bar Chart
How many loans defaulted?	Defaulted Loans	Defaulted Loans by Credit Score Band - Column Chart
What is the financial value of defaults?	Total Default Amount	Default Amount by Branch - Bar Chart
How severe is delinquency?	Average DPD	Average DPD by Risk Status - Column Chart
How are risk events changing over time?	Total Risk Events	Risk Events Trend by Month and Event Type - Line Chart

8.3.16 Risk Drivers

The page can use customer credit score, annual income, employment status, credit history and debt-to-income characteristics as explanatory drivers. Risk-event fields such as event type, event date, event days, event amount and risk status support deeper analysis than a single default flag.

8.3.17 Management Use

A manager can begin with the overall Default Rate and then determine whether the result is concentrated in a particular loan product, credit-score band or branch. This makes the KPI actionable because the visual explains the segment behind the headline number.

8.3.18 Dashboard Objective

The Collections & Recovery page focuses on what happened after amounts became due. Its question is not who defaulted, but how much should have been collected, how much was actually paid, how much payment delay exists, and how much has been recovered from risky exposures.

Analytical Question	KPI	Supporting Visual
How much was due?	Total Due Amount	Due Amount Trend - Line Chart
How much was actually paid?	Total Paid Amount	Due vs Paid by Month - Clustered Column
How effective is collection?	Collection Rate %	Collection Rate by Payment / Event Method - Bar Chart
What share of payments is late?	Late Payment Rate %	Late Payment Rate by Product - Bar Chart
How much was recovered?	Total Recovery Amount	Recovery Amount by Risk Status - Column Chart

8.3.19 Core Formula

Collection Rate %

Collection Rate = Total Paid Amount / Total Due Amount. The KPI provides the portfolio-level result, while the supporting chart explains which method, product, branch or period is contributing to the gap between due and paid amounts.

8.3.20 Management Use

The page allows collections teams to identify underperforming segments, compare payment methods, detect products with higher late-payment behavior, and evaluate how recovery activity is distributed across risk categories.

8.3.21 Dashboard Objective

The Branch & Loan Officer Performance page focuses on operational distribution. It does not repeat portfolio or default analysis already shown elsewhere. Instead, it asks where activity is concentrated and which organizational units or officers manage the largest lending and collection workload.

Analytical Question	KPI	Supporting Visual
How many branches are represented?	Total Branches	Loan Volume by Branch - Bar Chart
How many loan officers are represented?	Total Loan Officers	Loans by Loan Officer - Bar Chart
What is the average workload?	Loans per Officer	Top Loan Officers by Number of Loans - Bar Chart
How much lending is managed by officers?	Approved Amount per Officer	Approved Amount by Loan Officer - Bar Chart
How much collection activity occurs by branch?	Collections per Branch	Collection Amount by Branch - Column Chart

8.3.22 Organizational Context

The organization dimension can provide branch name, city, officer name, designation and experience-related attributes where available. These fields help interpret differences in workload and activity without creating a separate dimension for every organizational attribute.

8.3.23 Dashboard Story Completion

Together, the four dashboard pages create one non-repetitive sequence: how much the bank lent and where it is concentrated; what level of realized credit risk occurred and which segments explain it; how much became due, was paid, and was recovered; and where branch and officer workload is concentrated. The same Calendar and shared dimensions preserve filter context across the story without redefining KPIs.

9. Proposed System Architecture

9.1 Architecture Layers

Layer	Primary Responsibility	Output
Source Layer	Provide original customer, loan, product, organization, collateral, payment and risk data.	Raw input datasets
Bronze	Ingest and reconcile source data with minimal business transformation.	Traceable raw SQL tables
Silver	Clean, standardize, validate and prepare trusted records.	Validated domain tables
Gold	Reshape trusted data into business-ready dimensions and facts.	Six Gold analytical views
Power BI Model	Create Calendar, relationships and reusable semantic structure.	Dimensional semantic model
Measure Layer	Apply DAX definitions for portfolio, risk, collections and performance KPIs.	Reusable business measures
Presentation Layer	Build interactive dashboards, slicers and diagnostic visuals.	Four analytical dashboard pages
Decision Layer	Interpret findings and identify management attention areas.	Business insights and recommendations

9.2 Logical Flow

Architecture Flow

Source Data -> Bronze -> Silver -> Gold (6 Views) -> Power BI Data Model -> DAX Measures & KPIs -> Exploratory Analysis -> Dashboards -> Business Insights

9.3 Architectural Benefits

- Traceability from dashboard results back to source data.
- Isolation of raw data from business reporting.
- Reusable trusted data for multiple analyses.
- Reduced risk of duplicated fact aggregation.
- Clear ownership of SQL transformation versus Power BI semantic logic.
- Foundation for future predictive outputs without redesigning the entire reporting solution.

10. Tools and Technologies

Tool / Technology	Role in the Project	Why It Is Used
SQL Server	Data storage, layered preparation and analytical views	Supports controlled transformations, validation queries and repeatable data preparation.
SQL	Data profiling, cleaning, standardization, integration and Gold view creation	Places data transformation close to the data source and enables transparent reconciliation.
Power BI Desktop	Semantic modeling, DAX calculations and dashboard development	Provides interactive business intelligence and cross-filtered analytical reporting.
Power Query	Connection and light transformation within Power BI where required	Supports controlled ingestion and model preparation without replacing SQL cleaning logic.
DAX	Calendar table, measures, ratios and filter-aware KPI calculations	Allows reusable calculations that respond to report context.
Power BI Service	Optional publishing, sharing and refresh environment	Supports governed distribution when the project moves from desktop development to deployment.

10.1 Technology Selection Principle

The project keeps heavy data preparation in SQL and uses Power BI primarily for semantic modeling, measures and visualization. This separation makes the report easier to validate and reduces the amount of hidden transformation logic inside the BI file.

10.2 Development Simplicity

SQL code is kept as clear and explainable as possible while preserving the required business logic. This improves maintainability, makes project walkthroughs easier, and allows reviewers to understand the transformation path without relying on unnecessarily complex constructs.

11. Data / System Requirements

Requirement Type	Requirement
Data	All required customer, loan, product, organization, payment and risk keys must be available for integration.
Data Quality	Mandatory keys must be complete; categories and data types must be standardized before Gold.
Modeling	Dimensions must have unique keys and facts must retain their intended business grain.
Functional	Users must be able to filter results by relevant date, product, customer and organizational dimensions.
Analytical	KPIs must answer defined business questions and supporting charts must explain the result.
Validation	Counts and monetary totals must reconcile between SQL Gold and Power BI.
Performance	The semantic model should avoid unnecessary columns, ambiguous relationships and duplicate calculations.
Security	Access to customer-level or sensitive credit information should be controlled according to role and business need.
Documentation	KPI definitions, data lineage, assumptions and validation results must be documented.

11.1 Non-Functional Expectations

- Clarity: the model and dashboard should be understandable to both technical and business reviewers.
- Consistency: the same KPI should return the same result wherever it is used.
- Traceability: a reported number should be explainable through the Gold view and its source lineage.
- Maintainability: transformations and measures should be reusable rather than duplicated.
- Scalability: the design should support additional predictive outputs or reporting pages in future.

12. Expected Deliverables

Deliverable	Description	Status / Position
Bronze Layer	Raw SQL ingestion layer retaining source values and reconciliation checks.	Completed
Silver Layer	Cleaned, standardized and validated SQL layer.	Completed
Gold Layer	Business-ready analytical layer.	Completed
Six Gold Views	3 dimensions + 3 fact-oriented analytical views.	Completed
Gold Validation	Row counts, amount reconciliation and dimension-key checks.	Completed
Power BI Data Model	Six Gold views plus DAX Calendar and dimensional relationships.	Completed
KPI Catalog	Portfolio, credit risk, collections, recovery and organizational measures.	Defined / Implemented progressively
Analytical Questions	Question -> KPI -> Chart mapping for each dashboard page.	Defined
Dashboard 1	Loan Portfolio Overview.	In progress - KPI cards and portfolio visuals under final formatting
Dashboard 2	Credit Risk & Default Analysis.	In progress - credit-risk KPIs and diagnostic charts under final formatting
Dashboard 3	Collections & Recovery.	Planned next - approved Collections and Recovery design
Dashboard 4	Branch & Loan Officer Performance.	Planned - approved Branch and Loan Officer design
Project Report	Business, technical, analytical and validation documentation.	This document
Presentation	Final management storyline and project demonstration.	Final project stage

12.1 Deliverable Integration

The deliverables are not independent artifacts. The SQL layers produce the Gold views; the Gold views feed the Power BI model; the model supports reusable measures; the measures answer analytical questions; and the dashboards communicate those answers to business users. This dependency chain is essential to the credibility of the final reporting solution.

13. Evaluation and Testing

13.1 Validation Approach

Testing is performed at multiple stages rather than only after the dashboard is complete. Source-to-Bronze checks validate ingestion, Bronze-to-Silver checks validate cleaning rules, Silver-to-Gold checks validate transformation and joins, and SQL-to-Power BI checks confirm that the semantic model has not changed validated totals.

Validation Area	Test	Result / Expectation
Gold Loan Fact	Validated loan row count	30,000 loans
Gold Cashflow Fact	Validated event row count	97,642 cashflow events
Gold Risk Events Fact	Validated event row count	92,932 risk events
Amount Reconciliation	Compare transformed monetary totals with validated source/Silver totals	Zero amount differences reported
Dimension Keys	Check fact records for unresolved dimension keys	Zero missing dimension keys reported
Power BI Model	Check 1:* relationships and filter behavior	Must reproduce Gold totals under equivalent filters
KPI Logic	Compare DAX results with direct SQL calculations	Must reconcile before sign-off
Dashboard Interaction	Test slicers, cross-filtering and page behavior	No ambiguous or contradictory results

13.2 User Acceptance Testing

User acceptance testing focuses on whether the report answers the intended management questions, whether filters behave predictably, whether KPI definitions are understandable, and whether the visual hierarchy makes it easy to move from a summary value to its business driver. Technical correctness and business usability are both required for acceptance.

Validation Outcome

The reported Gold validation passed with the stated row counts, zero amount differences and zero missing dimension keys. This provides a strong baseline for Power BI reconciliation.

14. Innovation and Added Value

14.1 End-to-End Analytical Design

The main added value is the integration of data engineering, dimensional modeling and business analysis into one coherent workflow. The project does not start with dashboard charts. It first builds a governed data foundation, then validates the model, and only then translates business questions into KPIs and visuals.

14.2 Non-Repetitive Dashboard Story

Each dashboard has a distinct purpose. Portfolio metrics remain on the Portfolio page, historical default and DPD metrics remain on the Credit Risk page, due/paid/recovery metrics remain on the Collections page, and organizational workload metrics remain on the Branch & Officer page. This reduces visual noise and produces a clearer management narrative.

14.3 KPI-to-Driver Framework

The project explicitly links every major KPI to charts that explain it. This is more useful than displaying a high number of independent visuals because the user can move from a result to a possible business driver. For example, Default Rate is decomposed by product, credit score and branch; Collection Rate is decomposed by time, method and product.

14.4 Scalable Foundation for Predictive Analytics

The six-view Gold structure and dimensional Power BI model provide a natural extension point for future machine-learning results. Prediction outputs can be stored as a new analytical table and connected to the same customer, loan, product and organization context without replacing the descriptive reporting solution.

Competitive Advantage

The project can be presented as a risk decision-support architecture rather than only a Power BI dashboard: governed SQL layers, validated Gold views, dimensional modeling, analytical questions, KPI governance and a clear predictive roadmap.

15. Team Roles and Responsibilities

Define the role of each team member. Roles should be adjusted based on the project track and team size.

Team Member	Role	Responsibilities
Omnia Mamdouh	Final Reports	Consolidation, technical writing, evidence review, report formatting, and final submission quality assurance.
Nesma Saleh	Data Cleaning and Data Analyst	Data profiling, cleansing rules, standardization, Silver-layer preparation, and analytical validation.
Nedaa Mohamed	Data Analyst	Business-question analysis, KPI requirements, exploratory analysis, and interpretation of portfolio and risk results.
Maysoun Adel	Business Intelligence Analyst	Power BI semantic modeling, relationship design, DAX measures, and dashboard integration.
Aya Mamdouh	Data Visualization Specialist	Dashboard layout, visual selection, KPI-to-chart mapping, interaction design, and presentation consistency.

16. Ethical, Security, and Privacy Considerations

16.1 Data Confidentiality

Banking data may contain sensitive customer, financial and credit information. Access to detailed records should therefore be limited to users with a legitimate analytical or operational need. Reports intended for broader management audiences should use aggregated information where customer-level detail is not required.

Consideration	Recommended Control
Customer Privacy	Minimize exposure of directly identifying information in analytical views and presentation screenshots.
Access Control	Use role-appropriate access to data sources, Power BI workspaces and report pages.
Data Minimization	Include only fields required for the defined analytical purpose.
Secure Storage	Keep source files, SQL databases and exported reports in controlled environments.
Responsible Credit Analysis	Use credit attributes for defined risk analysis and avoid unsupported conclusions about individuals.
Bias Awareness	Review future predictive models for unequal error patterns and document feature choices.
Auditability	Document KPI logic, data lineage, quality rules and model assumptions.
Presentation Security	Use anonymized or non-identifying examples in external or academic presentations where possible.

16.2 Ethical Use of Analytics

Analytical results should support informed review rather than replace professional judgment without governance. Historical associations, such as higher default rates in a segment, should not be presented as proof that every customer in that segment will default. This distinction becomes even more important when predictive modeling is introduced in future phases.

16.3 Regulatory Caution

The current project should not be represented as a full regulatory IFRS 9 or Basel implementation unless all required regulatory data, assumptions, controls and validation processes are available. The analytical framework can support risk monitoring and future ECL-related extensions, but regulatory claims require a higher governance standard.

17. Expected Impact

17.1 Expected Impact

Impact Area	Expected Benefit
Portfolio Visibility	One consistent view of lending size, composition and approved exposure.
Risk Monitoring	Faster identification of default concentration, delinquency and risk-event trends.
Collections	Clear comparison of due versus paid amounts and collection efficiency.
Recovery	Improved visibility into recovery amounts and the risk categories generating them.
Organization	Transparent comparison of branch activity and loan-officer workload.
Data Quality	Reusable validated SQL layers reduce repeated cleaning and reconciliation effort.
Decision Support	KPI-to-driver analysis helps management focus on the segments behind headline results.
Scalability	Gold and semantic layers can support future predictive and ECL-related analytical outputs.

18. Conclusion

The Enterprise Banking Risk Management Analytics Solution demonstrates a complete analytical workflow from raw banking data to management-oriented insight. The project first establishes a controlled SQL foundation through Bronze, Silver and Gold layers. It then uses six business-ready Gold views to feed a dimensional Power BI model containing shared customer, product and organization dimensions plus separate loan, cashflow and risk-event facts.

The analytical design is centered on business questions rather than visual quantity. Portfolio size and composition are analyzed separately from historical default risk, collections and recovery, and organizational performance. Reusable KPIs and supporting visuals provide a clear path from what happened to where and why it happened.

The validated Bronze, Silver, and Gold pipeline, the six-view Galaxy model, and the current Power BI semantic model provide a reliable foundation for management reporting. The final validation confirms 30,000 loans, 97,642 cashflow events, and 92,932 risk events, with zero reported amount differences and zero missing dimension keys. The report deliberately separates portfolio, realized credit risk, collections and recovery, and organizational performance so that each KPI is explained once by a diagnostic chart. Predictive risk and ECL remain future extensions and will only be presented as implemented after formal validation and integration.

19. Future Work

19.1 Future Work

Future Extension	Purpose
Default Prediction	Classify loans or customers as likely/non-likely to default based on validated historical features.
Probability of Default (PD)	Estimate a probability score that can be aggregated by customer, product, branch or portfolio.
Customer Risk Segmentation	Group customers into interpretable risk profiles after evaluating cluster characteristics.
Payment Behavior Segmentation	Identify on-time, occasional-late, frequent-late and severely delinquent behavior patterns.
Explainable AI	Use feature importance and, where appropriate, SHAP-style explanations to show major risk drivers.
Early Warning View	Combine current performance, predicted risk and payment behavior to prioritize monitoring.
Recovery Modeling	Estimate recoverability or recovery segments for high-risk/defaulted exposures if sufficient data is available.
ECL Extension	Explore Expected Credit Loss using PD x LGD x EAD only when valid LGD/EAD inputs and governance are available.
ML-to-SQL-to-Power BI Integration	Persist prediction results in SQL and surface them in a predictive dashboard beside historical analytics.

19.2 Future-State Analytical Story

Descriptive -> Diagnostic -> Predictive -> Prescriptive

The current project explains what happened and which business dimensions are associated with the result. Future work can add what is likely to happen next and what action should be prioritized, while preserving the validated historical reporting layer as the foundation.

20. References and Project Resources

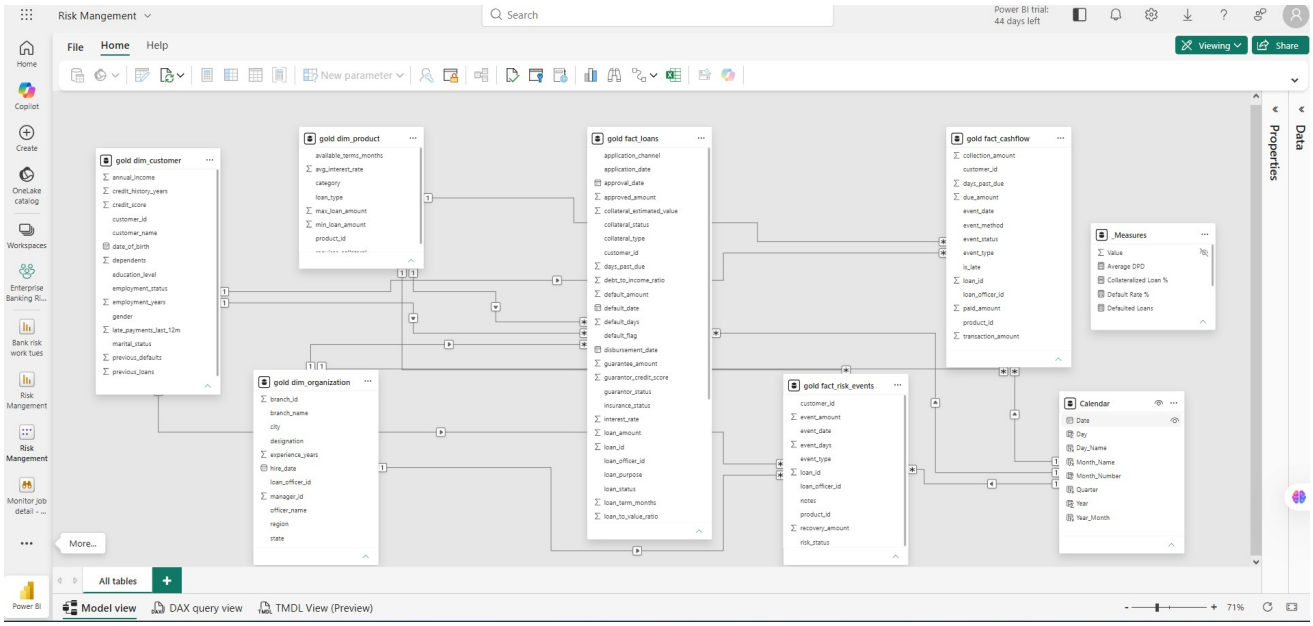
- Microsoft. Power BI documentation: data modeling, DAX, report design, and refresh guidance. Microsoft Learn. https://learn.microsoft.com/power-bi/
- Microsoft. SQL Server documentation: relational database design, views, data types, and query validation. Microsoft Learn. https://learn.microsoft.com/sql/sql-server/
- IFRS Foundation. IFRS 9 Financial Instruments. https://www.ifrs.org/issued-standards/list-of-standards/ifrs-9-financial-instruments/
- Basel Committee on Banking Supervision. Basel Framework. Bank for International Settlements. https://www.bis.org/basel_framework/
- Project resources: SQL scripts, validation queries, Power BI semantic model, KPI catalogue, analytical-question mapping, and implementation screenshots produced by the project team.

End of Report

Appendices

Appendix A: Power BI Data Model

Figure 3. Current Power BI dimensional model showing gold dim_customer, gold dim_product, gold dim_organization, gold fact_loans, gold fact_cashflow, gold fact_risk_events, Calendar, and _Measures.



Appendix B: Architecture Summary

Source Data -> Bronze -> Silver -> Gold (six views) -> Power BI semantic model -> DAX measures -> four management dashboards -> validated business insights. This flow preserves source traceability, separates business grains, and connects governed data preparation to decision-oriented reporting.

Item	Link
GitHub Repository	Not publicly available at the documentation date
Demo Video	Not publicly available at the documentation date
Prototype Link	Not publicly available at the documentation date
Dataset Link	Not publicly available at the documentation date
Dashboard Link	Not publicly available at the documentation date